

24. WRITE A C PROGRAM TO FIND THE VALUE OF Y USING

$$Y(x,n) = \begin{cases} 1+x & \text{where } n=1 \\ 1+x/n & \text{where } n=2 \\ 1+x^n & \text{when } n=3 \\ 1+nx & \text{when } n>3 \text{ or } n<1 \end{cases}$$

```
#include<stdio.h>
#include<math.h>
int main(){
    int x,n;
    float Y_function;
    printf("enter the value of x : ");
    scanf("%d",&x);
    printf("enter the value of n : ");
    scanf("%d",&n);
    if(n==1){
        Y_function=1+x;
        printf("the Y_function is 1+x and the value is %f",Y_function);
    }
    else if(n==2){
        Y_function=1+x/(n+0.0);
        printf("the Y_function is 1+x/n and the value is %f",Y_function);
    }
    else if(n==3){
        Y_function=1+pow(x,n);
        printf("the Y_function is 1+x^n and the value is %f",Y_function);
    }
    else if(n>3 || n<1){
        Y_function=1+n*x;
        printf("the Y_function is 1+nx and the value is %f",Y_function);
    }
    return 0;
}
```

```
> cd "c:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3\" ; if ($?) { gcc ASSIGNMENT_24.c -o ASSIGNMENT_24 } ; if ($?) { .\ASSIGNMENT_24 }
enter the value of x : 1
enter the value of n : 2
the Y_function is 1+x/n and the value is 1.500000
PS C:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3>
```

```
PS C> cd "c:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3\" ; if ($?) { gcc ASSIGNMENT_24.c -o ASSIGNMENT_24 } ; if ($?) { .\ASSIGNMENT_24 }
enter the value of x : 1
enter the value of n : 1
the Y_function is 1+x and the value is 2.000000
PS C:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3>
```

```
> cd "c:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3\" ; if ($?) { gcc ASSIGNMENT_24.c -o ASSIGNMENT_24 } ; if ($?) { .\ASSIGNMENT_24 }
enter the value of x : 2
enter the value of n : 3
the Y_function is 1+x^n and the value is 9.000000
PS C:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3>
```

```
> cd "c:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3\" ; if ($?) { gcc ASSIGNMENT_24.c -o ASSIGNMENT_24 } ; if ($?) { .\ASSIGNMENT_24 }
enter the value of x : 4
enter the value of n : 4
the Y_function is 1+nx and the value is 17.000000
PS C:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3>
```

25. PROGRAM TO PRINT TABLE OF ANY NO.

```
#include<stdio.h>

int main(){

    int n;

    printf("enter the number whose table u want to print : ");

    scanf("%d",&n);

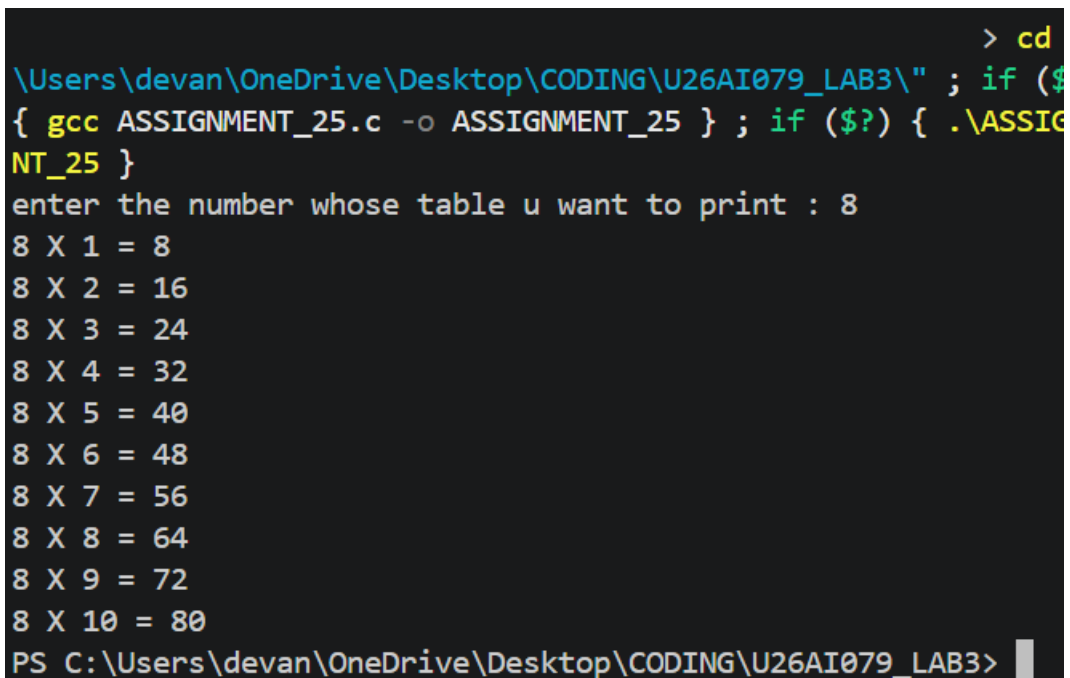
    for(int i=1;i<=10;i++){

        printf("%d X %d = %d\n",n,i,n*i);

    }

    return 0;

}
```



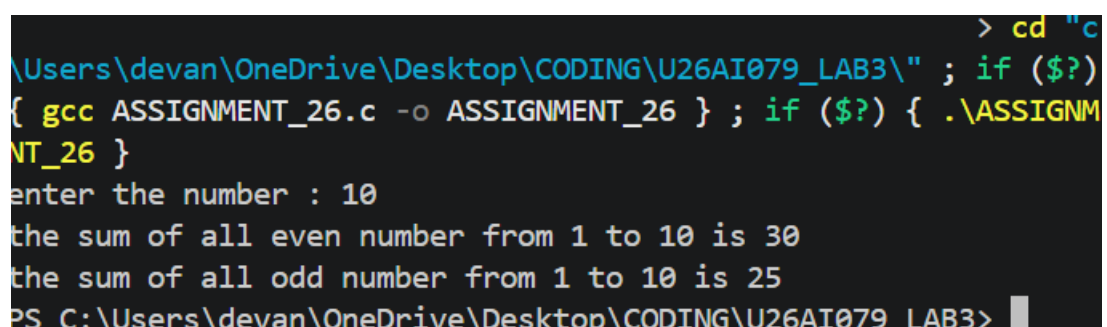
The screenshot shows a terminal window with a dark background. At the top right, a prompt '>' is followed by 'cd'. Below this, a command is entered: '\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3\" ; if (\$?){ gcc ASSIGNMENT_25.c -o ASSIGNMENT_25 } ; if (\$?) { .\ASSIGNMENT_25 }'. The program then prompts 'enter the number whose table u want to print : 8'. It then prints the multiplication table for 8, from 8 X 1 = 8 to 8 X 10 = 80. At the bottom, the prompt 'PS C:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3>' is visible.

```
> cd
\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3\" ; if ($?){ gcc ASSIGNMENT_25.c -o ASSIGNMENT_25 } ; if ($?) { .\ASSIGNMENT_25 }
enter the number whose table u want to print : 8
8 X 1 = 8
8 X 2 = 16
8 X 3 = 24
8 X 4 = 32
8 X 5 = 40
8 X 6 = 48
8 X 7 = 56
8 X 8 = 64
8 X 9 = 72
8 X 10 = 80
PS C:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3>
```

26. PROGRAM TO PRINT SUM OF ALL ODD AND EVEN NO B/W 1 TO N.

```
#include<stdio.h>

int main(){
    int n,odd_sum=0,even_sum=0;
    printf("enter the number : ");
    scanf("%d",&n);
    for(int i=1;i<=n;i++){
        if(i%2==0) even_sum+=i;
        else odd_sum+=i;
    }
    printf("the sum of all even number from 1 to %d is %d\n",n,even_sum);
    printf("the sum of all odd number from 1 to %d is %d",n,odd_sum);
    return 0;
}
```



```
> cd "C:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3\" ; if ($?) { gcc ASSIGNMENT_26.c -o ASSIGNMENT_26 } ; if ($?) { .\ASSIGNMENT_26 }
enter the number : 10
the sum of all even number from 1 to 10 is 30
the sum of all odd number from 1 to 10 is 25
PS C:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3>
```

27. PROGRAM TO CHECK WHETHER A NO IS PRIME OR NOT.

```
#include<stdio.h>

int main(){
    int n,count=0;

    printf("enter the number you want to check prime or not : ");

    scanf("%d",&n);

    for(int i=1;i<=n;i++){
        if(n%i==0) count++;
    }

    if(count==2) printf("the entered number %d is prime",n);

    else printf("the entered number %d is not prime",n);

    return 0;
}
```

```
> cd "c:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3\" ; if ($?)
{ gcc ASSIGNMENT_27.c -o ASSIGNMENT_27 } ; if ($?) { .\ASSIGNME
NT_27 }
enter the number you want to check prime or not : 12
the entered number 12 is not prime
PS C:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3>
```

```
> cd "c:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3\" ; if ($?)
{ gcc ASSIGNMENT_27.c -o ASSIGNMENT_27 } ; if ($?) { .\ASSIGNME
NT_27 }
enter the number you want to check prime or not : 7
the entered number 7 is prime
PS C:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3>
```

28. PROGRAM TO CHECK WHETHER A NO IS ARMSTRONG OR NOT.

```
#include<stdio.h>

#include<math.h>

int main(){

    int n,q=0,temp=n,checker=0,count=0;

    printf("enter the number you want to check ARMSTRONG or not : ");

    scanf("%d",&n);

    while(n>0){

        count++;

        n=n/10;

    }

    n=temp;

    while(n>0){

        q=n%10;

        checker+=pow(q,count);

        n=n/10;

    }

    if(temp==checker) printf("the given number is ARMSTRONG");

    else printf("the given number is not ARMSTRONG");

    return 0;

}
```

```
> cd "c:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3\" ; if ($?) { gcc ASSIGNMENT_28.c -o ASSIGNMENT_28 } ; if ($?) { .\ASSIGNMENT_28 }
enter the number you want to check ARMSTRONG or not : 153
the given number is ARMSTRONG
PS C:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3>
```

```
> cd "c:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3\" ; if ($?) { gcc ASSIGNMENT_28.c -o ASSIGNMENT_28 } ; if ($?) { .\ASSIGNMENT_28 }
enter the number you want to check ARMSTRONG or not : 12
the given number is not ARMSTRONG
PS C:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3>
```

29. PROGRAM TO CHECK WHETHER A NO IS PALINDROM OR NOT.

```
#include<stdio.h>

int main(){
    int n,q,reverse=0;
    printf("enter the Number : ");
    scanf("%d",&n);
    int temp;
    temp=n;
    while(n>0){
        q=n%10;
        reverse+=q;
        reverse=reverse*10;
        n=n/10;
    }
    if(temp==reverse/10) printf("%d is a PALINDROM",temp);
    else printf("%d is not a PALINDROM",temp);
    return 0;
}
```

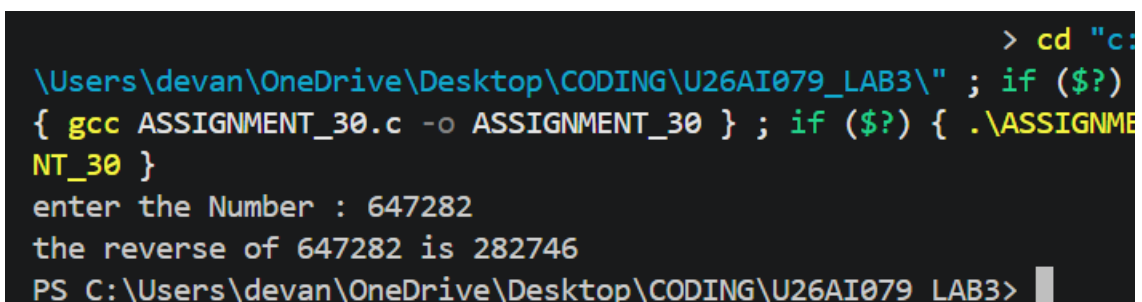
```
> cd "c:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3\" ; if ($?) {
{ gcc ASSIGNMENT_29.c -o ASSIGNMENT_29 } ; if ($?) { .\ASSIGNME
NT_29 }
enter the Number : 12121
12121 is a PALINDROM
PS C:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3>
```

```
> cd "c:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3\" ; if ($?) {
{ gcc ASSIGNMENT_29.c -o ASSIGNMENT_29 } ; if ($?) { .\ASSIGNME
NT_29 }
enter the Number : 38788
38788 is not a PALINDROM
PS C:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3>
```

30. PROGRAM TO REVERSE OF A GIVEN NO.

```
#include<stdio.h>

int main(){
    int n,q,reverse=0;
    printf("enter the Number : ");
    scanf("%d",&n);
    int temp;
    temp=n;
    while(n>0){
        q=n%10;
        reverse+=q;
        reverse=reverse*10;
        n=n/10;
    }
    printf("the reverse of %d is %d",temp,reverse/10);
    return 0;
}
```

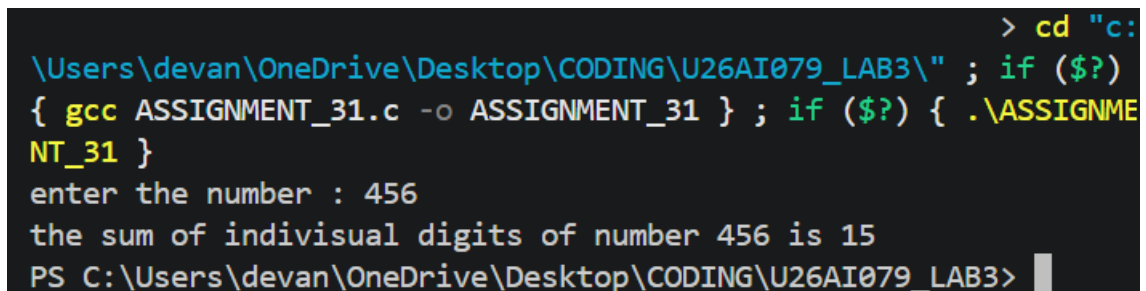


```
> cd "c:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3\"
\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3\" ; if ($?) { gcc ASSIGNMENT_30.c -o ASSIGNMENT_30 } ; if ($?) { .\ASSIGNMENT_30 }
enter the Number : 647282
the reverse of 647282 is 282746
PS C:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3>
```

31. PROGRAM TO PRINT SUM OF INDIVIDUAL DIGITS OF A 'N' DIGIT NO.

```
#include<stdio.h>

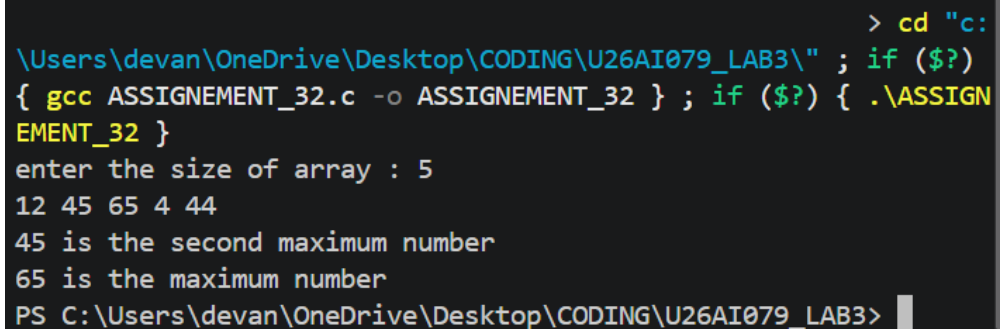
int main(){
    int n,sum=0;
    printf("enter the number : ");
    scanf("%d",&n);
    int temp=n;
    while(n>0){
        int quotient=n%10;
        sum=sum+quotient;
        n=n/10;
    }
    printf("the sum of indivisual digits of number %d is %d",temp,sum);
    return 0;
}
```



```
> cd "c:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3\" ; if ($?) { gcc ASSIGNMENT_31.c -o ASSIGNMENT_31 } ; if ($?) { .\ASSIGNMENT_31 }
enter the number : 456
the sum of indivisual digits of number 456 is 15
PS C:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3>
```


32. PROGRAM TO FINDOUT MAX AND SECOND MAX FROM 'N' NOS .

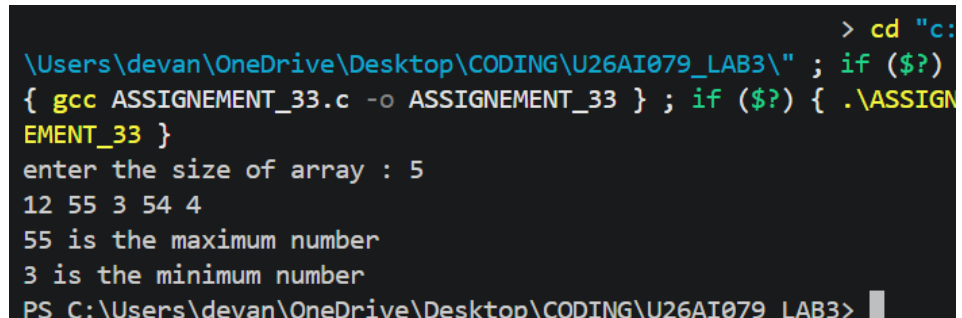
```
#include<stdio.h>
int main(){
    int n;
    printf("enter the size of array : ");
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<=n-1;i++){
        scanf("%d",&arr[i]);
    }
    for(int i=0;i<=n-1;i++){
        int count=0;
        for(int j=0;j<=n-1;j++){
            if(arr[i]>=arr[j]) count++;
        }
        if(count==n-1) printf("%d is the second maximum number \n",arr[i]);
        if(count==n){
            printf("%d is the maximum number ",arr[i]);
        }
    }
    return 0;
}
```



```
> cd "c:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3\" ; if ($?) { gcc ASSIGNMENT_32.c -o ASSIGNMENT_32 } ; if ($?) { .\ASSIGNMENT_32 }
enter the size of array : 5
12 45 65 4 44
45 is the second maximum number
65 is the maximum number
PS C:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3>
```

33. PROGRAM TO FINDOUT MAX AND MIN FROM 'N' NOS.

```
#include<stdio.h>
int main(){
    int n;
    printf("enter the size of array : ");
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<=n-1;i++){
        scanf("%d",&arr[i]);
    }
    for(int i=0;i<=n-1;i++){
        int countmax=0;
        int countmin=0;
        for(int j=0;j<=n-1;j++){
            if(arr[i]>=arr[j]) countmax++;
            if(arr[i]<=arr[j]) countmin++;
        }
        if(countmax==n) printf("%d is the maximum number \n",arr[i]);
        if(countmin==n){
            printf("%d is the minimum number \n",arr[i]);
        }
    }
}
return 0;
```



```
> cd "c:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3\" ; if ($?) {
gcc ASSIGNMENT_33.c -o ASSIGNMENT_33 } ; if ($?) { .\ASSIGNMENT_33 }
enter the size of array : 5
12 55 3 54 4
55 is the maximum number
3 is the minimum number
PS C:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3>
```

34. PROGRAM TO PRINT FIBONACCI SERIES.(0,1,1,2,3...)

```
#include<stdio.h>
```

```
int main(){
```

```
    int n;
```

```
    printf("enter the number of terms in fibonacci series : ");
```

```
    scanf("%d",&n);
```

```
    int x=0;
```

```
    int y=1;
```

```
    printf("%d ",x);
```

```
    printf("%d ",y);
```

```
    int count=1;
```

```
    while(count<=n-2){
```

```
        int sum=x+y;
```

```
        x=y;
```

```
        y=sum;
```

```
        printf("%d ",sum);
```

```
        count++;
```

```
    }
```

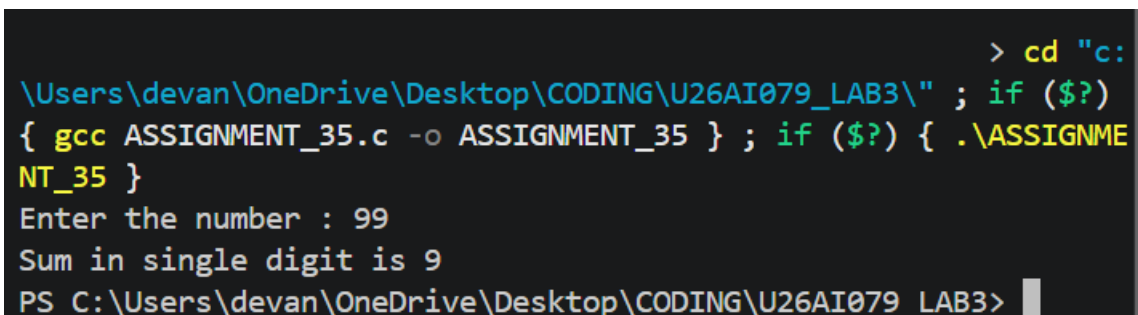
```
}
```

```
> cd "c:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3\" ; if ($?)  
{ gcc ASSIGNMENT_34.C -o ASSIGNMENT_34 } ; if ($?) { .\ASSIGNME  
NT_34 }  
enter the number of terms in fibonacci series : 10  
0 1 1 2 3 5 8 13 21 34  
PS C:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3>
```

35. READ A NUMBER N AND PRINT A SINGLE DIGIT ANSWER SHOWING SUM OF DIGIT OF N.

```
#include<stdio.h>

int main(){
    int n,q;
    printf("Enter the number : ");
    scanf("%d",&n);
    int sum=0;
    while(n>=10){
        int sum_temp=0;
        while(n!=0){
            q=n%10;
            sum_temp+=q;
            n=n/10;
        }
        n=sum_temp;
        sum=sum_temp;
    }
    printf("Sum in single digit is %d",sum);
}
```



```
> cd "c:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3\" ; if ($?) { gcc ASSIGNMENT_35.c -o ASSIGNMENT_35 } ; if ($?) { .\ASSIGNMENT_35 }
Enter the number : 99
Sum in single digit is 9
PS C:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3>
```

36. PROGRAM TO PRINT 1,3,5,7,9.....N.

```
#include<stdio.h>

int main(){

    int n;

    printf("Enter the Number Of Terms in the A.P : ");

    scanf("%d",&n);

    int a=1,d=2,l;

    l=a+(n-1)*d;

    for(int i=a;i<=l;i=i+d){

        printf("%d ",i);

    }

}
```

```
> cd "c:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3\" ; if ($?) {
gcc tempCodeRunnerFile.c -o tempCodeRunnerFile } ; if ($?) {
.\tempCodeRunnerFile }
Enter the Number Of Terms in the A.P : 10
1 3 5 7 9 11 13 15 17 19
PS C:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3>
```

37. PROGRAM TO PRINT 2,4,6,8,10,12.....N.

```
#include<stdio.h>

int main(){
    int n;

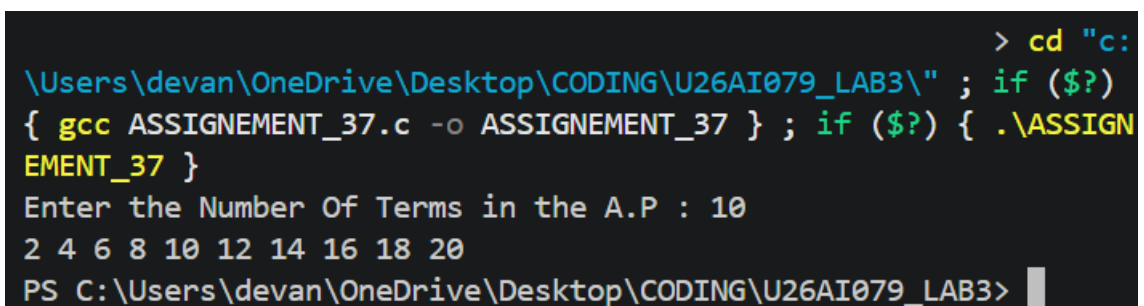
    printf("Enter the Number Of Terms in the A.P : ");

    scanf("%d",&n);

    int a=2,d=2,l;

    l=a+(n-1)*d;

    for(int i=a;i<=l;i=i+d){
        printf("%d ",i);
    }
}
```



```
> cd "c:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3\" ; if ($?) { gcc ASSIGNMENT_37.c -o ASSIGNMENT_37 } ; if ($?) { .\ASSIGNMENT_37 }
Enter the Number Of Terms in the A.P : 10
2 4 6 8 10 12 14 16 18 20
PS C:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3>
```

38. PROGRAM TO PRINT 1,4,9,16,25,.....N.

```
#include<stdio.h>

int main(){
    int n;

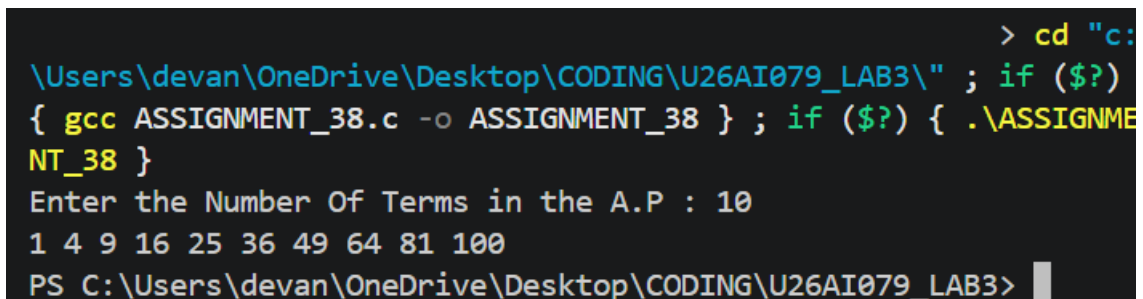
    printf("Enter the Number Of Terms in the A.P : ");

    scanf("%d",&n);

    int a=1,d=1,l;

    l=n;

    for(int i=a;i<=l;i=i+d){
        printf("%d ",i*i);
    }
}
```



```
> cd "c:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3\" ; if ($?) { gcc ASSIGNMENT_38.c -o ASSIGNMENT_38 } ; if ($?) { .\ASSIGNMENT_38 }
Enter the Number Of Terms in the A.P : 10
1 4 9 16 25 36 49 64 81 100
PS C:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3>
```

39. PROGRAM TO PRINT $1/1!+2/2!+3/3!+4/4!,\dots,N.$

```
#include<stdio.h>

int main(){
    int n;

    printf("Enter the Number Of Terms : ");

    scanf("%d",&n);

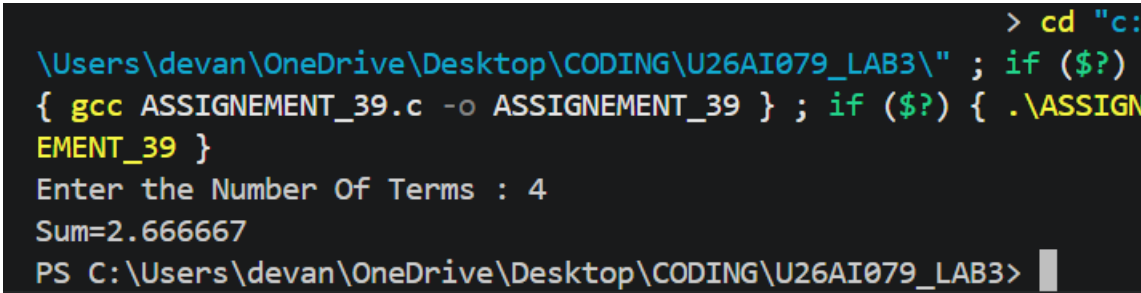
    float sum=0;

    for(int i=1;i<=n;i++){
        int fact=1;

        for(int j=1;j<=i;j++){
            fact=fact*j;
        }

        sum=sum+(i/(fact+0.0));
    }

    printf("Sum=%f",sum);
}
```



```
> cd "C:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3\" ; if ($?) {
{ gcc ASSIGNMENT_39.c -o ASSIGNMENT_39 } ; if ($?) { .\ASSIGNMENT_39 }
Enter the Number Of Terms : 4
Sum=2.666667
PS C:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3>
```


40. READ NUMBERS TILL A NEGATIVE NUMBER IS ENTERED AND CALCULATE SUM OF A LIST OF NUMBERS READ.

```
#include<stdio.h>

int main(){

    int n;

    int sum=0;

    do{

        printf("Enter the number : ");

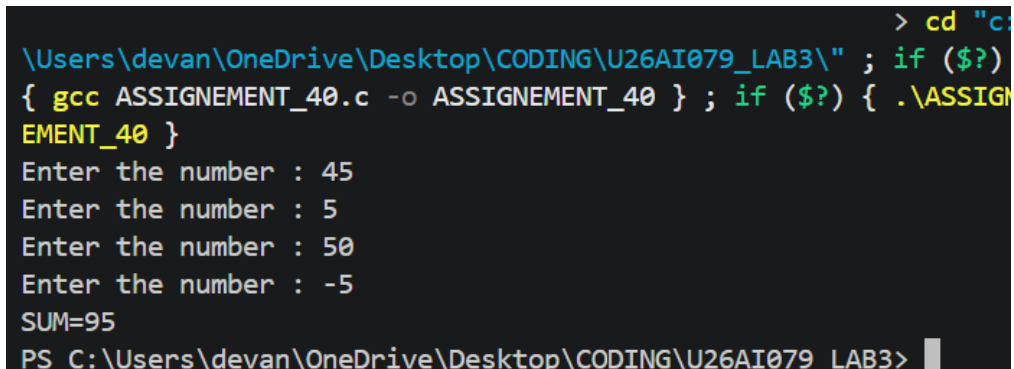
        scanf("%d",&n);

        sum=sum+n;

    }while(n>0);

    printf("SUM=%d",sum);

}
```



```
> cd "C:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3\"
{ gcc ASSIGNMENT_40.c -o ASSIGNMENT_40 } ; if ($?) { .\ASSIGNMENT_40 }
Enter the number : 45
Enter the number : 5
Enter the number : 50
Enter the number : -5
SUM=95
PS C:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3>
```

41. READ A NUMBER N AND PRINT FACTOR OF N.(EX 28= 1,2,4,7,14,28)

```
#include<stdio.h>
```

```
int main(){
```

```
    int n;
```

```
    printf("Enter the Number : ");
```

```
    scanf("%d",&n);
```

```
    for(int i=1;i<=n;i++){
```

```
        if(n%i==0) printf("%d ",i);
```

```
    }
```

```
}
```

```
> cd "c:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3\" ; if ($?) { gcc ASSUGNEMENT_41.c -o ASSUGNEMENT_41 } ; if ($?) { .\ASSUGNEMENT_41 }
Enter the Number : 28
1 2 4 7 14 28
PS C:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3>
```

28) 42. READ A NUMBER N AND PRINT IF N IS PRIME OR COMPOSITE.

```
#include<stdio.h>

int main(){

    int n,count=0;

    printf("Enter the Number : ");

    scanf("%d",&n);

    for(int i=1;i<=n;i++){

        if(n%i==0) count++;

    }

    if(count==2) printf("The entered number which is %d is Prime",n);

    else printf("The entered number which is %d is composite",n);

}
```

```
> cd "c:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3\" ; if ($?) { gcc ASSIGNMENT_42.c -o ASSIGNMENT_42 } ; if ($?) { .\ASSIGNMENT_42 }
Enter the Number : 4
The entered number which is 4 is composite
PS C:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3>
```

```
> cd "c:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3\" ; if ($?) { gcc ASSIGNMENT_42.c -o ASSIGNMENT_42 } ; if ($?) { .\ASSIGNMENT_42 }
Enter the Number : 11
The entered number which is 11 is Prime
PS C:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3>
```

43. WRITE A C PROGRAM TO CALCULATE THE COMMISSION FOR A SALES REPRESENTATIVE AS PER THE SALES AMOUNT GIVEN BELOW.

If sales<=Rs. 500, commission is 5%
If sales> but <=2000, commission is Rs. 35 plus 10% above Rs. 500
If sales>2000 but <=5000, commission is Rs. 185 plus 12% above Rs. 2000
If sales>5000, commission is 12.5%

```
#include<stdio.h>
```

```
int main(){
```

```
    int sales;
```

```
    printf("Enter the sales amount : ");
```

```
    scanf("%d",&sales);
```

```
    if(sales<=500) printf("Commission is %f",(5*sales)/100.0);
```

```
    else if(sales>500 && sales<=2000) printf("Commission is %f",35+(10*(sales-500))/100.0);
```

```
    else if(sales>2000 && sales<=5000) printf("Commission is %f",185+(12*(sales-2000))/100.0);
```

```
    else printf("Commission is %f",((12.5)*sales)/100.0);
```

```
}
```

```
> cd "c:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3\" ; if ($?) { gcc ASSIGNMENT_43.c -o ASSIGNMENT_43 } ; if ($?) { .\ASSIGNMENT_43 }  
Enter the sales amount : 200  
Commission is 10.000000  
PS C:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3>
```

```
> cd "c:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3\" ; if ($?) { gcc ASSIGNMENT_43.c -o ASSIGNMENT_43 } ; if ($?) { .\ASSIGNMENT_43 }  
Enter the sales amount : 1500  
Commission is 135.000000  
PS C:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3>
```

```
> cd "c:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3\" ; if ($?) { gcc ASSIGNMENT_43.c -o ASSIGNMENT_43 } ; if ($?) { .\ASSIGNMENT_43 }  
Enter the sales amount : 2500  
Commission is 245.000000  
PS C:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3>
```

```
> cd "c:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3\" ; if ($?) { gcc ASSIGNMENT_43.c -o ASSIGNMENT_43 } ; if ($?) { .\ASSIGNMENT_43 }  
Enter the sales amount : 6000  
Commission is 750.000000  
PS C:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3>
```

44. WRITE A PROGRAM IN C FOR THE FOLLOWING.
AN ELECTRIC DISTRIBUTION COMPANY CHARGES ITS DOMESTIC CONSUMERS AS
FOLLOWS.

Consumption in unit	Rate for Charge
0-200	Re 0.50 per unit
210-400	Rs. 100 plus Re 0.65 per unit excess of 200
401-600	Rs. 230 plus Re 0.80 per unit excess of 400
Above 600	Rs. 425 plus Rs. 125 per unit excess of 600

Print the amount to be paid by the consumer.

```
#include<stdio.h>
```

```
int main(){
```

```
    int consump_unit;
```

```
    printf("Enter the the Consumption In Unit : ");
```

```
    scanf("%d",&consump_unit);
```

```
    if(consump_unit<=200) printf("The Amount To Be Paid is %f",0.5*consump_unit);
```

```
    else if(consump_unit>200 && consump_unit<=400) printf("The Amount To Be Paid is  
%f",100+0.65*(consump_unit-200));
```

```
    else if(consump_unit>400 && consump_unit<=600) printf("The Amount To Be Paid is  
%f",230+0.8*(consump_unit-400));
```

```
    else printf("The Amount To Be Paid is %d",425+125*(consump_unit-600));
```

```
}
```

```
> cd "c:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3\" ; if ($?) { gcc ASSIGNMENT_44.c -o ASSIGNMENT_44 } ; if ($?) { .\ASSIGNMENT_44 }  
Enter the the Consumption In Unit : 100  
The Amount To Be Paid is 50.000000  
PS C:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3>
```

```
> cd "c:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3\" ; if ($?) { gcc ASSIGNMENT_44.c -o ASSIGNMENT_44 } ; if ($?) { .\ASSIGNMENT_44 }  
Enter the the Consumption In Unit : 250  
The Amount To Be Paid is 132.500000  
PS C:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3>
```

```
> cd "c:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3\" ; if ($?) { gcc ASSIGNMENT_44.c -o ASSIGNMENT_44 } ; if ($?) { .\ASSIGNMENT_44 }  
Enter the the Consumption In Unit : 450  
The Amount To Be Paid is 270.000000  
PS C:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3>
```

```
> cd "c:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3\" ; if ($?) { gcc ASSIGNMENT_44.c -o ASSIGNMENT_44 } ; if ($?) { .\ASSIGNMENT_44 }  
Enter the the Consumption In Unit : 610  
The Amount To Be Paid is 1675  
PS C:\Users\devan\OneDrive\Desktop\CODING\U26AI079_LAB3>
```